

Gustavo Rodriguez

Mechanical Engineering Student | Expected Graduation: December 2028
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Research & Engineering Leadership

IEEE EXO, University of South Florida

EXO Project Lead / Chair

Tampa, FL

August 2025 – Present

- Lead a multidisciplinary lower-limb exoskeleton program spanning mechanical design, electrical, controls, research, fabrication, testing, safety, procurement, and technical documentation.
- Define requirements for a unilateral hip/knee assistive prototype using AK80-9 actuators, IMUs, encoders, Jetson Orin Nano, and Teensy 4.1; perform OpenSim gait comparisons across healthy, hemiparetic, and SHIA datasets to support assistance timing, torque targets, and validation planning.
- Coordinate EXO chair responsibilities with the IEEE board, including technical/member events, workshops, recruiting, career-fair and employer outreach, sponsor/industry communication, budgets, meeting logistics, and cross-team scheduling.

CARRT Lab, University of South Florida

REU Research Assistant

Tampa, FL

August 2026 – Present

- Analyze IMU and Vicon motion-capture data and support experimental setup, participant data collection, sensor validation, post-processing, data-quality review, and technical documentation for human-motion research.
- Develop a modular ESP32/IMU sensor for rugby lineout motion capture; designed a SolidWorks ankle/thigh enclosure reduced from 80 x 60 mm to 43 x 32 mm (71.3% smaller footprint), with ventilation and integrated adjustable strap slots to avoid reliance on athletic tape or sleeves. Helped validate the PCB and supported compact PCB/layout integration.

Engineering Experience

Lonza

Reliability Engineering Intern

Tampa, FL

August 2026 – December 2026

- Investigate recurring Xcelodose 600S and microbalance failures through physical inspection, mechanical alignment, spring/load-cell geometry, corner-load behavior, SOP-to-practice comparison, 6M, Gemba, and failure-mode analysis.

Lonza

Facilities, Engineering & Maintenance Intern

Tampa, FL

May 2026 – August 2026

- Led a Lean/5S asset-identification improvement that reduced representative equipment search time from about 9 minutes to 1.5 minutes (83%).
- Built Power Apps/Power Automate/SharePoint workflows for maintenance and manufacturing traceability, including guided request intake and structured run/status tracking.

Technical Skills & Credentials

Engineering: SolidWorks 2025, CAD/assembly modeling, drawing/specification review, FDM 3D printing, RCA/FMEA, equipment validation

Software / Hardware: MATLAB, OpenSim, Python, Excel, Power Apps, Power Automate, SharePoint, Arduino/ESP32, Jetson/Teensy, IMUs, sensors and actuators

PCB / Electronics: Altium PCB Design coursework and Altium Education certificate in progress; IEEE Altium PCB design workshops; PCB validation/integration support

Credentials: Lean Six Sigma Yellow Belt; CSWA CAD Design; CSWA Sustainability; ReCF Pre-Engineering & Robotics; CITI Biomedical / Research Integrity

Selected Technical Projects

SMURF – Maintenance Request System | Power Apps, Power Automate, SharePoint

- Built a two-app maintenance workflow with structured condition questions, automated urgency determination, HTML notifications/reminders, assignment/status tracking, and searchable records.

Mime Robotic Hand | Python, MediaPipe, ESP32, PCA9685, Servo Control

- Integrated computer vision with an open-source 5-DOF tendon-driven hand; developed calibration, mapping, smoothing/hysteresis, reverse-servo logic, and interactive control modes.

Education

University of South Florida

B.S. Mechanical Engineering, GPA: 3.33

Tampa, FL
Expected December 2028

Coursework: Dynamics, Mechanics of Solids, Thermodynamics, Materials Engineering, Electrical Systems, CAD & Engineering, PCB Design / Altium.